



Joshua Kenneth Robertson Tabh

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Overview: I am a natural scientist with interests in evolution, physiology, mathematics, and biodiversity conservation. So far, I have sought to fuse these interests by studying the mechanisms that species have evolved to persist in dynamic environments (both thermal and otherwise). This work relies heavily on interdisciplinarity, and involves rethinking how we model evolution and species-environment interactions. Outside of research, I advocate for science-based conservation action, currently by serving as a board member for the Nova Scotia Nature Trust.

Languages: English proficient, Swedish intermediate.

Computing languages: R, C++, Python, Julia, Bash, Stan

1 | Positions

- Sven and Lily Lawksi Postdoctoral Fellow (2024–Current). Lund University, Lund, Sweden.
Supervisors: Drs. Andreas Nord and Cecilia Lundberg
- Wenner-Gren Postdoctoral Fellow (2022–2024). Lund University, Lund, Sweden.
Supervisor: Dr. Andreas Nord
- Parental leave (2021–2022).
Supervisor: Between May of 2021 and April of 2022, I provided full-time parental care for my child, born March of 2021.
- Research associate (2021). Brock University, Saint Catharines, Canada.
Supervisor: Dr. Glenn Tattersall

2 | Education

- PhD Environmental and Life Sciences (2016 – 2021). Trent University, Peterborough, Canada.
Topic: Interactions between thermal and psychosocial stress in vertebrates.
Title: Surface temperature regulation during stress exposure (6 chapters, 319 pages).
Supervisors: Drs. Gary Burness (Trent University) and Gabriela Mastromonaco (Toronto Zoo).
- MSc Biology (2014–2016). McMaster University, Hamilton, Canada.
Title: Reproductive competition and success in a joint-nesting cuckoo (4 chapters, 142 pages).
Supervisor: Dr. James Quinn (McMaster University).
- Honours BSc Zoology (2008 – 2012). University of Guelph, Guelph, Canada.
Honours Title: Synergistic antagonism of Wnt signalling by Dkk1 and Nkd1.
Supervisor: Dr. Terry Van Raay (Guelph University).

3 | Grants and scholarships

Total funding acquired to date (at current currency conversion rates): 393,627.00 CAD

· Kungl. Fysiografiska Sällskapet i Lund – Travel endowment	Value: 15,000 SEK
· Carl Tryggers Postdoktorstipendier	Value: 855,150 SEK
· Sven och Lily Lawksi Postdoktorstipendier	Value: 672,000 SEK
· Stiftelsen Lunds Djurskyddsfonden (2023).	Value: 25,000 SEK
· Stiftelsen Längmanska Kulturfonden (2023).	Value: 25,000 SEK
· Stiftelsen Lars Hieras Minne (2023).	Value: 30,000 SEK
· Kungl. Fysiografiska Sällskapet i Lund – Jan Löfqvist Endowment (2022).	Value: 96,000 SEK
· Wenner-Gren Postdoktorstipendier (2022-2024).	Value: 672,000 SEK
· Dean's PhD Scholarship Trent University (2016-2020).	Value: 16,000 CAD
· Queen Elizabeth II Graduate Scholarship Government of Ontario (2016). Offered but declined.	Value: 15,000 CAD
· WOS Research Grant Wilson Ornithological Society (2016).	Value: 1,500 USD
· Queen Elizabeth II Graduate Scholarship Government of Ontario (2015).	Value: 15,000 CAD
· Graduate Entrance Award - McMaster University (2014).	Value: 3,000 CAD

4 | Board Memberships

- Nova Scotia Nature Trust Executive Board (2023–Current). Halifax, Nova Scotia, Canada.
- Eastern Loggerhead Shrike Recovery Team (2016–2019). Toronto, Ontario, Canada.

5 | Publications

Total number of peer-reviewed publications: 20. Number of first author publications: 12.

Articles in review or in press:

- (1) **Tabh, J.K.R.**, Nyquist, P., Nord, A. (2026). Environmental effects overtake selection to shape avian body size. *Science*. In review.
- (2) Ryding, S., Klaassen, M., **Tabh, J.K.R.**, Tattersall, G.J., Nord, A., & Symonds, M. (2026). Beyond single traits: predicting species' phenotypic responses to climatic warming. *Nature Biodiversity*, In review.
- (3) **Tabh, J.K.R.** (2026). Robustness: unpacking priorities and possible futures. *Current Biology*. In press (dispatch).

Peer-reviewed:

- (1) **Tabh, J.K.R.**, Persson, E., Correia, M., Ó Cuív, C., Thorall, E., & Nord, A. (2025). Limited evidence that body size shrinking and shape-shifting alleviate thermoregulatory pressures in a warmer world. *Communications biology*, 8(1), 1–13.
- (2) Grieves L., Hing J.S., **Tabh J.K.R.**, Quinn J.S.. (2025). Offspring sex ratio in a communal breeding bird is male-biased when pre-breeding rainfall is low. *Journal of Avian Biology*, e03262.
- (3) [†]Stewart, N.D., [†]**Tabh, J.K.R.** & Burness, G. (2024). Island living indirectly affects hair glucocorticoid levels in a small mammal. *Ecological and Evolutionary Physiology*, 97(5), 263–273.
- (4) ^{★†}Nord, A., [†]Persson, E., [†]**Tabh, J.K.R.**, & [†]Thorall, E. (2024). Shrinking body size may not provide meaningful thermoregulatory benefits in a warmer world. *Nature Ecology and Evolution*, 1–3.
- (5) **Tabh, J.K.R.**, Hartjes, M., & Burness, G. (2023). Endotherms trade body temperature regulation for the stress response. *Proceedings of the Royal Society B*, 290(2010), 20231251.
- (6) **Tabh, J.K.R.** & Nord, A. (2023). Temperature-dependent developmental plasticity and its effects on Allen's and Bergmann's rules in endotherms. *Integrative and Comparative Biology*, 63(3), 758–771.

- (7) Penney, C. M., **Tabh, J.K.R.**, Wilson, C.C., & Burness, G. (2022). Within-generation and transgenerational plasticity of a temperate salmonid in response to thermal acclimation and acute temperature stress. *Physiological and Biochemical Zoology*, 95(6), 484–499.
- (8) **Tabh, J.K.R.**, Mastromonaco GF, Burness G. (2022). Stress-induced changes in body surface temperature are repeatable, but do not differ between urban and rural birds. *Oecologia*, 198(3), 663–677.
- (9) Tapper S., **Tabh, J.K.R.**, Tattersall G.J., Burness G. (2021). Changes in body surface temperature play an under-appreciated role in the avian immune response. *Physiological and Biochemical Zoology* 95(2), 152–167.
- (10) **Tabh, J.K.R.**, Burness G., Wearing O., Tattersall G.J., Mastromonaco G.F. (2021). Infrared thermography as a technique to measure physiological stress in birds: Body region and image angle matter. *Physiological Reports*, 9(11):e14865. doi: 10.14814/phy2.14865.
- (11) Kummrow, M.S., Pimm, R.H., Mackie, P. M., **Tabh, J.K.R.**, & Mastromonaco, G.F. (2021). Fecal adrenal hormone patterns during ovulatory and non-ovulatory reproductive cycles in female veiled chameleons (*Chamaeleo calyptratus*). *General and Comparative Endocrinology*, 310, 113822.
- (12) Penney, C.M., Burness, G., **Tabh, J.K.R.**, & Wilson, C. C. (2021). Limited transgenerational effects of environmental temperatures on thermal performance of a cold-adapted salmonid. *Conservation Physiology*, 9(1), coab021.
- (13) **Tabh, J.K.R.** (2021). Surface temperature regulation during stress exposure: evidence of a trade-off between thermoregulation and the physiological stress response in birds. *PhD Thesis*, Trent University, Canada. url [here](#).
- (14) **Robertson, J.K.**, Mastromonaco, G.F., & Burness, G. (2020). Social hierarchy reveals thermoregulatory trade-offs in response to repeated stressors. *Journal of Experimental Biology*, 223(21), jeb229047.
- (15) ***Robertson, J. K.**, Mastromonaco, G., & Burness, G. (2020). Evidence that stress-induced changes in surface temperature serve a thermoregulatory function. *Journal of Experimental Biology*, 223(4), jeb213421.
- (16) **Robertson, J. K.**, Caldwell, J.R., Grieves, L.A., Samuelsen, A., Schmaltz, G.S., & Quinn, J.S. (2018). Male parental effort predicts reproductive contribution in the joint-nesting, Smooth-billed Ani (*Crotophaga ani*). *Journal of Ornithology*, 159, 471–481.
- (17) **Robertson, J.K.**, Muir, C., Hurd, C.S., Hing, J.S., & Quinn, J.S. (2017). The effect of social group size on feather corticosterone in the co-operatively breeding Smooth-billed Ani (*Crotophaga ani*): An assay validation and analysis of extreme social living. *PLoS One*, 12(3), e0174650.
- (18) **Robertson JK** et al. (2016). Reproductive competition and success in a joint-nesting cuckoo. *MSc Thesis*, McMaster University, Canada. url [here](#).
- (19) **Robertson J.K.** et al. (2014). Targeting the Wnt pathway in zebrafish as a screening method to identify novel therapeutic compounds. *Experimental Biology and Medicine*, 239(2), 169–176.
- (20) Lum, W.M., **Robertson, J.K.**, & Van Raay, T.J. (2011). Dishevelled2 is a stable protein during early zebrafish development. *Zebrafish*, 8(2), 65–71.

★ Highlighted in popular media (Inside JEBTM; <https://bit.ly/3DOBkCV>) and selected as issue cover article

† Authors contributed equally.

6 | Invited and public talks (by date)

- (1) **Tabh, J.K.R.** (2025). Bergmann, Urbanek, and Cope: The evolution of body size and the role of physiology. Department of Biology, St. Francis Xavier University, Antigonish, Nova Scotia, Canada.
- (2) **Tabh, J.K.R.** (2025). Joel, Carl, Adam, and the appeal to antiquity: Does old wisdom really help us explain why the shape of extant birds is changing? European Ornithological Union, Bangor, Wales.

- (3) **Tabh J.K.R.** (2023) – Understanding Allen – Identifying the proximate drivers of Allen’s Rule. Lund University, Tissue Development and Evolution Group, Lund, Sweden.
- (4) **Tabh J.K.R.** (2020) – Stressed birds may not be hot, but they save well. McMaster University, Hamilton, Ontario, Canada.
- (5) **Tabh J.K.R.** (2019) – The complicated matters of stress and body temperature. Lund University, Lund, Sweden.
- (6) **Tabh J.K.R.** (2018) – Non-invasive measures of stress physiology in birds. Annual Ontario Bird Banding Association, Cayuga, Canada.
- (7) **Tabh J.K.R.** (2017) – Stress in non-human animals: measurement and meaning for conservation biology. East Elgin Secondary School, Aylmer, Canada.

7 | Conference symposia and chairing

- (1) *Symposium*. Adapting to the world ahead – how the developmental environment shapes adult phenotypes for better or worse (2025). Designed together with Elin Persson (Lund University) and Dr. Elisa Thorall (La Rochelle Université). European Ornithological Union, Bangor, Wales.
- (2) *Chairing*. Oral presentations. European Ornithological Union, Bangor, Wales (2025).
- (3) *Chairing*. Oral presentations. Changing Animals in a Changing World Meeting, Lund, Sweden. (2025).

8 | Conference presentations (by date)

- (1) **Tabh, J.K.R.** (2025). Joel, Carl, Adam, and the appeal to antiquity. *European Ornithological Society Meeting*. Bangor, Wales.
- (2) **Tabh, J.K.R.** (2024). Body size in a warmer world: lessons from population ecology. *Changing Animals in a Changing World Meeting*. Lund, Sweden.
- (3) **Tabh, J.K.R.** (2024). Climate change may not drive body size declines in birds. *Nordic Society Oikos*. Lund, Sweden.
- (4) Penney CM, **Tabh J.K.R.**, Burness G, Wilson C. (2021). Within- and transgenerational plasticity of a temperate salmonid in response to thermal acclimation and acute temperature stress. *Society for Experimental Biology*. Online, United Kingdom
- (5) **Robertson J.K.** (2020) – Dominant birds are hot but not bothered. *Comparative Physiology and Biochemistry Workshop*. Rice Lake, Ontario, Canada.
- (6) **Robertson J.K.** (2019) – Stress-induced peripheral hypothermia: role of the sympathetic nervous system in avian thermal modulation. *Society for Integrative and Comparative Biology*. Tampa, Florida, USA.
- (7) **Robertson J.K.** (2018) – Running cold: what thermal profiles reveal about stress in birds. *FLIR InfraCanada*. Niagara, Ontario, Canada.
- (8) **Robertson J.K.** (2018) – The heat is on: what thermal profiles can tell us about stress in birds. *Canadian Association of Zoos and Aquariums*. Cambridge, Ontario, Canada.
- (9) **Robertson J.K.** (2017) – Stress in non-human animals: measurement and meaning for conservation and research. *Canadian Association of Zoos and Aquariums*. Ottawa, Ontario, Canada.
- (10) **Robertson J.K.**, Caldwell JR, Grieves L, Samuelsen A, Schmaltz G, Quinn JS. (2016) – Reproductive skew in a joint-nesting cuckoo: does parental effort pay off? *Animal Behavior*, Columbia, Missouri, USA.
- (11) **Robertson J.K.**, Caldwell JR, Grieves L, Samuelsen A, Schmaltz G, Quinn JS. (2016) – Does parental effort correlate with reproductive contribution in communal nests? Patterns of reproductive skew in a social cuckoo. *Canadian Society of Zoologists*, London, Ontario, Canada.

9 | Supervision, mentorship and training

- Dr. Chloe Kaczvinsky, Postdoctoral fellow (2025–current). Lund University, Lund, Sweden.

- *Elin Persson, PhD (2024–current). Lund University, Lund, Sweden.
- Marina Jiménez, MSc (2022–2023). Lund University, Lund, Sweden.
- Micheal Campbell, MSc (2021). Trent University, Peterborough, Canada.
- *Mariah Hartjes, BSc (2020). Trent University, Peterborough, Canada.
- Avneesh Sachdev, BSc (2019). McMaster University, Hamilton, Canada.
- Lianne Ralph, MSc (2018). Trent University, Peterborough, Canada.
- Alexandra Tekatch, BSc (2018). McMaster University, Hamilton, Canada.
- *Conner Hurd, BSc (2016). McMaster University, Hamilton, Canada.
- *Jing Sheng Hing, Meghan Healey, MSc (2015). McMaster University, Hamilton, Canada.
Training occurred in Cabo Rojo, Puerto Rico, USA.
- *Albin Bjärhall, Kristie Stein, Sheena Wheeler, BSc. (2014). Training occurred in Cabo Rojo, Puerto Rico, USA.

* Mentorship led to peer-reviewed publications.

10 | Peer reviewing

I provide regular peer reviews for several academic journals including:

Current Biology, Functional Ecology, Oikos, Oecologia, Journal of Experimental Biology, Journal of Thermal Biology, Biological Journal of the Linnean Society, Conservation Physiology, Canadian Journal of Zoology, Journal of Avian Research, PLOS One, Journal of Marine Biology, Ecography, Journal of Animal Ecology, and Ecology Letters.

11 | Academic teaching and guest lecturing

I regularly provide lectures for undergraduate courses in statistics, ecology, and physiology. Where possible, I have also sought to provide lectures for courses outside of the science community (i.e. in arts and sciences). Some courses I have provided lectures for include (but are not limited to):

- Animal physiology. Trent University, Canada: BIOL-3830
- Principles of the nervous system. Trent University, Canada: BIOL-3210
- Neuroendocrinology Trent University, Canada: BIOL-4840
- Ecological genetics. Trent University, Canada: BIOL-3600
- Conservation biology. McMaster University, Canada: BIOL-4AA3
- Research design and data analysis. Trent University, Canada: BIOL-4030
- Role of zoos and aquariums in conservation. University of Toronto: BIOC-62H3
- Case studies in arts and science. Trent University, Canada: ASCI-2001
- Interdisciplinary perspectives in arts and sciences. Trent University, Canada: ASCI-1001

Beyond lecturing, I have offered numerous teaching assistantships, whenever available to me. In some cases (indicated with an asterisk below), this involved constructing seminar lectures and laboratory projects. A short list of courses I have offered teaching assistantships for is as follows:

- Foundations of cellular and molecular biology. Trent University, Canada: BIOL-1030
- Cellular and molecular biology. McMaster University, Canada: BIOL-1A03
- Biodiversity, evolution and humanity. McMaster University, Canada: BIOL-1M03
- Foundations of biodiversity Trent University, Canada: BIOL-1020
- *Principles of the nervous system. Trent University, Canada: BIOL-3210
- Ornithology. Trent University, Canada: BIOL-4140H
- *Research design and data analysis. Trent University, Canada: BIOL-4030
- *Neuroendocrinology Trent University, Canada: BIOL-4840
- Conservation biology. McMaster University, Canada: BIOL-4AA3
- *Ecological genetics. Trent University, Canada: BIOL-3600

12 | Education for community

Helping to advancing mathematical and statistical knowledge across disciplines is deeply important to me. To this end, I co-launched weekly statistical learning meetings while at my doctoral institution (Trent University). These meetings, known as “Data Lunches”, were attended by early science learners, government employees, and later-career academics. Beyond this, I am an active member of the InterBayes research network, based out of Uppsala University and led by Måns Magnusson (Stan developer and Associate Professor in Statistics). The purpose of this network is promote the application of Bayesian philosophy and statistical tools across disciplines.

13 | Management/oversight experience

- (1) I organised two, 4-month research expeditions to the Cabo Rojo National Wildlife Preserve (Cabo Rojo, Puerto Rico) across 2014–2015 with the objective of studying the physiology and cooperative behaviour of a highly social bird species (the Smooth-billed Ani). Expeditions required the independent training and supervision of 6 research personnel (3/expedition), planning of data collection, management of research finances, and management of field research equipment (including a field research vehicle).
- (2) During my doctorate, I designed and orchestrated a 4-month, experimental research expedition throughout southern Ontario, Canada, with the goal of testing trade-offs between thermoregulation and the stress-response in a model bird (the Black-capped Chickadee). This included development and construction of an avian holding facility (Cayuga, Ontario, Canada), training and management of one research assistant, and execution of data collection (i.e. capture, blood sampling, leg-ringing, and remote, infra-red thermography).

14 | Public and industry outreach

Communicating knowledge to the public, industry, and governments is a critical aspect of publicly-funded research. To achieve this, I have provided public blog-posts (e.g. via <https://yukonwildlife.ca/news> and <https://tattersalllab.com/category/ramphastos-ramblings/>), internationally-available interviews (e.g. <https://shorturl.at/oqtw4>; <https://www.trentu.ca/about/news-events/38527>), nature tours for children, young female STEM learners and adults (Cayuga, Ontario, Canada, and Cabo Rojo National Wildlife Preserve, Cabo Rojo, Puerto Rico), and guest lecture for all ages.

To engage industry with relevant technical and practical interests (e.g. animal physiology and remote-physiological sensing), I have provided presentations at industry-centred conferences seldom attended by the academic community (including the Canadian Association of Zoos and Aquariums Annual Meeting, and FLIR InfraCanada; section 7). Such presentations have led to the establishment of ongoing connections between my research program and an industry partner (the Yukon Wildlife Preserve; see above).